

Jing Liu 刘璟

Senior Researcher
Systems Research Group
Microsoft Research Asia

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RESEARCH INTERESTS

I am a computer systems researcher specializing in operating systems, storage systems, and distributed systems. My current research particularly focuses on the CPU-side systems stack for agentic AI and the co-evolution of next-generation computing systems and models. I pursue holistic, full-stack designs in which in-situ code generation can eliminate cross-layer overheads and indirections to attain maximal end-to-end efficiency.

EDUCATION

Department of Computer Sciences, University of Wisconsin – Madison

Ph.D. in Computer Sciences

Aug. 2024

M.S. in Computer Sciences

May 2022

Advisors: Prof. Andrea Arpaci-Dusseau & Prof. Remzi Arpaci-Dusseau

Thesis: *Scale, Performance, and Fault Tolerance in a Filesystem Semi-Microkernel*

Software Institute, Nanjing University

B.E. in Software Engineering Rank: 1/238

May 2016

Thesis: *Depth-aware layered edge for elastic RGB-D object proposal*

PROFESSIONAL EXPERIENCE

Microsoft Research Asia

Oct. 2024 – Present

Senior Researcher.

Google, Madison, WI

Summer 2019

Software Engineering Intern. Mentor: Dr. Marc De Kruijf

Microsoft Research, Redmond, WA

Summer 2018

Research Intern. Mentor: Dr. Irene Zhang

HONORS AND AWARDS

Erik Riedel Best Paper Award, USENIX Conference on File and Storage Technologies (FAST)

2025

Selected as one of 2 best papers from 167 submissions; top 1.2%

Meta PhD Research Fellowship, Meta Inc.

2022 – 2024

Sole awardee in the distributed systems research area worldwide

Computer Sciences Department Golden Brick Award, UW-Madison

2022

CS Summer Research Award, UW-Madison

2017

Graduate Research Fellowship, UW-Madison

2016 – 2017

Outstanding Undergraduate Thesis, University Highest Honor, Jiangsu Provincial Award

2016

National Scholarship, Ministry of Education of China

2013

PUBLICATIONS

[†] Student mentored by me.

Refereed Conference Publications

- [1] Shawn (Wanxiang) Zhong, Junxuan Liao, **Jing Liu**, Mai Zheng, Andrea C. Arpaci-Dusseau, and Remzi H. Arpaci-Dusseau. *Don't Let AI Agents YOLO Your Files: Information and Control in Agent-Native Filesystems*. In Proceedings of the 32nd ACM Symposium on Operating Systems Principles (SOSP '26), Prague, Czechia, September 2026. ([arXiv:2604.13536](#))
- [2] Yaoqi Chen, Jinkai Zhang, Baotong Lu, Qianxi Zhang, Chengruidong Zhang, **Jing Liu**, Jingjia Luo, Di Liu, Huiqiang Jiang, Qi Chen, Bailu Ding, Xiao Yan, Jiawei Jiang, Chen Chen, Mingxing Zhang, Yuqing Yang, Fan Yang, and Mao Yang. *RetroInfer: A Vector-Storage Approach for Scalable Long-Context LLM Inference*. In Proceedings of the 52nd International Conference on Very Large Data Bases (VLDB '26), August 2026. (Covered by [Hugging Face Trending Papers](#) [[url](#)] and [AI Research Roundup](#) [[url](#)])
- [3] Zhongjie Chen[†], Wentao Zhang, Yulong Tang[†], Ran Shu, Fengyuan Ren, Tianyin Xu, and **Jing Liu**. *Xkernel: Principled Performance Tunability of Operating System Kernels*. In Proceedings of the 20th USENIX Symposium on Operating Systems Design and Implementation (OSDI '26), July 2026.
- [4] Chuandong Li, Ran Yi, Zonghao Zhang, **Jing Liu**, Changwoo Min, Jie Zhang, Yingwei Luo, Xiaolin Wang, Zhenlin Wang, and Diyu Zhou. *Aeolia: A Fast and Secure Userspace Interrupt-Based Storage Stack*. In Proceedings of the 31st ACM Symposium on Operating Systems Principles (SOSP '25), pages 479–495. ACM, October 2025.
- [5] Shawn (Wanxiang) Zhong[†], **Jing Liu**, Andrea Arpaci-Dusseau, and Remzi Arpaci-Dusseau. *Revealing the Unstable Foundations of eBPF-Based Kernel Extensions*. In Proceedings of the 20th European Conference on Computer Systems (EuroSys '25), pages 21–41. ACM, March 2025.
- [6] **Jing Liu**, Yifan Dai, Andrea C. Arpaci-Dusseau, and Remzi H. Arpaci-Dusseau. *Fast, Transparent Filesystem Microkernel Recovery with Ananke*. In Proceedings of the 23rd USENIX Conference on File and Storage Technologies (FAST '25), pages 1–18. USENIX Association, February 2025. (**Erik Riedel Best Paper Award**)
- [7] Yifan Dai, **Jing Liu**, Andrea Arpaci-Dusseau, and Remzi Arpaci-Dusseau. *Symbiosis: The Art of Application and Kernel Cache Cooperation*. In Proceedings of the 22nd USENIX Conference on File and Storage Technologies (FAST '24), pages 51–69. USENIX Association, February 2024.
- [8] **Jing Liu**, Anthony Rebello, Yifan Dai, Chenhao Ye, Sudarsun Kannan, Andrea C. Arpaci-Dusseau, and Remzi H. Arpaci-Dusseau. *Scale and Performance in a Filesystem Semi-Microkernel*. In Proceedings of the 28th ACM Symposium on Operating Systems Principles (SOSP '21), pages 819–835. ACM, October 2021.
- [9] Irene Zhang, Amanda Raybuck, Pratyush Patel, Kirk Olynyk, Jacob Nelson, Omar S Navarro Leija, Ashlie Martinez, **Jing Liu**, Anna Kornfeld Simpson, Sujay Jayakar, et al. *The Demikernel Datapath OS Architecture for Microsecond-scale Datacenter Systems*. In Proceedings of the 28th ACM Symposium on Operating Systems Principles (SOSP '21), pages 195–211. ACM, October 2021.
- [10] Suli Yang, **Jing Liu**, Andrea C. Arpaci-Dusseau, and Remzi H. Arpaci-Dusseau. *Principled Schedulability Analysis for Distributed Storage Systems using Thread Architecture Models*. In Proceedings of the 13th USENIX Symposium on Operating Systems Design and Implementation (OSDI '18), pages 161–176. USENIX Association, October 2018.
- [11] Ramnatthan Alagappan, Aishwarya Ganesan, **Jing Liu**, Andrea Arpaci-Dusseau, and Remzi Arpaci-Dusseau. *Fault-Tolerance, Fast and Slow: Exploiting Failure Asynchrony in Distributed Systems*. In Proceedings of the 13th USENIX Symposium on Operating Systems Design and Implementation (OSDI '18), pages 390–408. USENIX Association, October 2018.
- [12] **Jing Liu**, Tongwei Ren, Bingkun Bao, and Jia Bei. *Depth-aware Layered Edge for Object Proposal*. In Proceedings of the IEEE International Conference on Multimedia and Expo (ICME '16), pages 1–6. IEEE, July 2016.
- [13] **Jing Liu**, Tongwei Ren, and Jia Bei. *Elastic Edge Boxes for Object Proposal on RGB-D images*. In Proceedings of International Conference on Multimedia Modeling (MMM '16), Lecture Notes in Computer Science, pages 199–211. Springer International Publishing, 2016.

arXiv Preprints

- [1] Zihan Zhao[†], Baotong Lu, Shengjie Lin[†], Yizou Chen, **Jing Liu**, Yanqi Zhang, Ziming Miao, Ming-Chang Yang, Haiying Shen, Qi Chen, and Fan Yang. *Unifying Sparse Attention with Hierarchical Memory for Scalable Long-Context LLM Serving*. arXiv Preprint, 2026. ([arXiv:2604.26837](#))

[2] Yuming Xu, Qianxi Zhang, Qi Chen, Baotong Lu, Menghao Li, Philip Adams, Mingqin Li, Zengzhong Li, **Jing Liu**, Cheng Li, and Fan Yang. *Scalable Distributed Vector Search via Accuracy Preserving Index Construction*. arXiv Preprint, 2025. ([arXiv:2512.17264](https://arxiv.org/abs/2512.17264)) (Covered by Microsoft Research Focus [[url](#)])

Refereed Journals and Workshops

[1] Inho Choi, Anand Bonde, **Jing Liu**, Joshua Fried, Irene Zhang, and Jialin Li. *ML-native Dataplane Operating Systems*. In Proceedings of the 16th ACM SIGOPS Asia-Pacific Workshop on Systems (APSys '25), pages 8–14. ACM, October 2025.

[2] **Jing Liu**, Xiangpeng Hao, Andrea Arpaci-Dusseau, Remzi Arpaci-Dusseau, and Tej Chajed. *Shadow Filesystems: Recovering from Filesystem Runtime Errors via Robust Alternative Execution*. In Proceedings of the 16th ACM Workshop on Hot Topics in Storage and File Systems (HotStorage '24), pages 15–22. ACM, 2024.

[3] Suli Yang, **Jing Liu**, Andrea Arpaci-Dusseau, and Remzi Arpaci-Dusseau. *Principled Schedulability Analysis for Distributed Storage Systems Using Thread Architecture Models*. ACM Transactions on Storage, 19(2):1–47, May 2023.

[4] **Jing Liu**, Andrea C. Arpaci-Dusseau, Remzi H. Arpaci-Dusseau, and Sudarsun Kannan. *File Systems as Processes*. In 11th USENIX Workshop on Hot Topics in Storage and File Systems (HotStorage '19). USENIX Association, July 2019.

[5] Irene Zhang, **Jing Liu**, Amanda Austin, Michael Lowell Roberts, and Anirudh Badam. *I'm Not Dead Yet! The Role of the Operating System in a Kernel-Bypass Era*. In Proceedings of the Workshop on Hot Topics in Operating Systems (HOTOS '19), pages 73–80. ACM, May 2019.

[6] **Jing Liu**, Tongwei Ren, Yuantian Wang, Sheng-Hua Zhong, Jia Bei, and Shengchao Chen. *Object proposal on RGB-D images via elastic edge boxes*. Neurocomputing, 236:134–146, May 2017.

SERVICE

Shadow PC Co-chair, EuroSys '26

Program Committee: SOSP '25, APSys '25, NSDI '26, OSDI '26, SOSP '26, FAST '27, HOTOS '27

External Review Committee: ATC '24

Artifact Evaluation Committee: ATC/OSDI '22

Topic (OS & Emerging Hardware) Moderator, SOSP '21 Diversity Workshop

President, W-ACM, UW Madison Chapter of ACM's Women in Computing Aug. 2021 – Aug. 2022

PRESENTATIONS AND INVITED TALKS

Systems Research: What We Build, Why We Build, and How We Do It?

Invited talk at MSRA Interns Event, Beijing, China. Aug. 2025

Invited talk at MSRA – PKU Open Day, Beijing, China. Jul. 2025

Invited talk at MSRA Open Day to Students, Beijing, China. Jun. 2025

Fast, Transparent Filesystem Microkernel Recovery with Ananke

Talk at FAST '25, Santa Clara, CA. Feb. 2025

Scale and Performance in a Filesystem Semi-Microkernel.

Invited talk at UIUC, Champaign, IL. Oct. 2022

Invited talk at ENS de Lyon – France, virtual. Mar. 2022

Invited talk at Samsung MSL Tech Talk, virtual. Jun. 2021

Talk at SOSP '21, virtual. Oct. 2021

File Systems as Processes.

Talk at HotStorage '19, Renton, WA. Jul. 2019

Elastic Edge Boxes for Object Proposal on RGB-D Images.

Talk at MMM '16, Miami, MI. Jan. 2016

TEACHING EXPERIENCE

Computer Sciences Department of UW–Madison, WI <i>Instructor</i> Class: Introduction to Operating Systems.	Summer 2023
Computer Sciences Department of UW–Madison, WI <i>Teaching Assistant</i> Class: Distributed Systems. Instructor: Prof. Andrea Arpaci-Dusseau	Spring 2022, Spring 2023
Computer Sciences Department of UW–Madison, WI <i>Teaching Assistant</i> Class: Advanced Operating System. Instructor: Prof. Andrea Arpaci-Dusseau	Spring 2018
Software Institute of Nanjing University, Nanjing, China <i>Teaching Assistant, and Head Teaching Assistant</i> Class: Computer Operating System. Instructors: Prof. Bin Luo & Prof. Jidong Ge	Spring 2015, Spring 2016

MENTORING

<i>A General Inference Engine for Sparse Attention</i> Zihan Zhao, University of Virginia, Ph.D. student Shengjie Lin, Georgia Tech University, Master student	May 2025 – Aug. 2026 Aug. 2025 – Aug. 2026
<i>Principled Performance Tunability in Operating System Kernels</i> Zhongjie Chen, Tsinghua University, Ph.D. student Yulong Tang, Renmin University of China, Master student	Apr. 2025 – July. 2026 Sep. 2025 – May. 2026
<i>Smart Caching in Vector Storage Systems</i> Sukjoon Oh, KAIST, Ph.D. student	Oct. 2024 – Present
<i>Unstable Foundations of eBPF Programs (EuroSys '25)</i> Shawn Zhong, UW–Madison, Ph.D. student	Jun. 2024 – May 2025
<i>DiskANN Case Study for Application Reaction to Filesystem Failure</i> Luo Zhong Zhou, UW–Madison, Undergraduate	May 2023 – Nov. 2023